

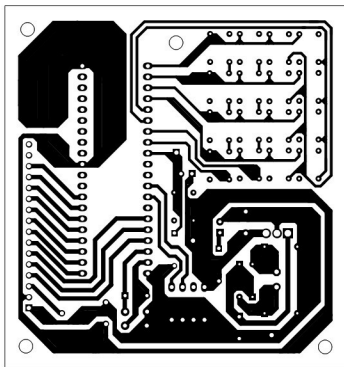


Fig. 2: An actual-size, single-side PCB for the circuit

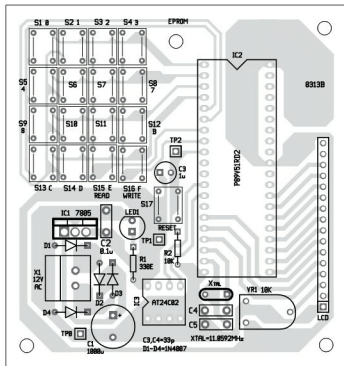


Fig. 3: Component layout for the PCB

5. Send chip address byte (say, A1H) along with read/write '1' bit ('0' for write or '1' for read) followed by an acknowledge bit

6. Read data from the EEPROM followed by an acknowledge bit

7. Send stop condition

Working of the circuit is simple. You can save data at any location on the EEPROM using the user interface and read it again later. The data is

never lost even if the power fails.

Construction and testing

An actual-size, single-side PCB for the circuit is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC base for the microcontroller and the EEPROM. Before inserting the IC, check the supply voltage. Using preset VR1 set the LCD contrast for proper display.

Test points are given in the table for checking. When the circuit is switched on, the 'EFY EEPROM test' message is displayed on the first line of the LCD and 'e read f save' is displayed on the second line of the LCD. Press 'F' key if you want to save data in the EEPROM. Message 'save your data' is displayed on the LCD and then it prompts you to enter the address location where you want to

save the data. You can enter the address, say, '22' (or any other address from 00 to FE) of the memory location. Then 'enter the data' message appears. You can enter any data, say, 34, and press 'E' to save the data or 'F' not to save the data.

Now switch off the power and turn it back 'on.' Press 'E' key to read the data. Message 'read your data' appears on the LCD. Thereafter, the

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

LCD display shows 'enter the address' message for entering the address (say, '22') of the memory location. The LCD display now will show 'stored data is: 34,' where '34' is the data you have entered in the above step.

Software program

The source program for the microcontroller is written in 'C' language and compiled using the Keil μ Vision4 compiler. The generated hex code is burnt into the microcontroller by using a suitable programmer. The program is well commented and easy to understand. When power is switched on or reset switch S17 is pressed momentarily, the microcontroller executes the main program.

First, it initialises the LCD and executes 'while' loop. 'EFY EEPROM test' and 'e read f save' messages are displayed on the first and second lines of the LCD module, respectively. If all the columns of the switches are high, key_scan() function checks for pressing of 'E' and 'F' keys. If 'E' key is pressed, read_module() function executes. If 'F' key is pressed, save_module() function executes. Function read_module() displays corresponding messages and finally reads the data from EEPROM using i2c_read() function. Reading of data from the EEPROM follows I²C protocol as stated above.

Similarly, save_module() function displays corresponding messages and finally saves the data in the EEPROM using i2c_write() function. Writing data to the EEPROM follows I²C protocol as stated above.

The i2c.h, lcd.h and keyboard.h header files are used for I²C protocol communication, LCD mode and 4×4 matrix keypad in the source program. ●

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